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Internalization of Ionic Transport Ability of Polymer Semiconductors via Photochemical Cross-Linking

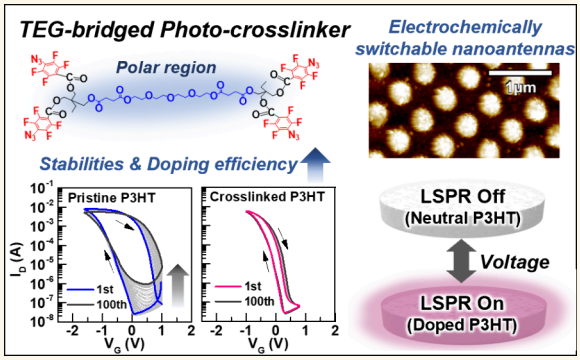
Won Jun Pyo, Seunghyun Lee, Syed Zahid Hassan, Dowan Kim, Junho Jung, Evan S. H. Kang,* and Dae Sung Chung*

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ABSTRACT: In the field of organic electronics and optics, there is rapidly growing interest in enhancing both charge transport and the ion transport properties of semiconductors, particularly in light of recent emerging technologies such as organic electrochemical transistors (OECTs) and switchable organic nanoantennas. Herein, we propose a universal method for internalizing the ionic transport properties of conventional polymer semiconductors. The incorporation of a tetrafluorophenyl azide-based photochemical cross-linker with a tetraethylene glycol bridge into poly(3-hexylthiophene) (P3HT) significantly enhances the performance and operational stability of ion-gating devices. Changes in the characteristics of the OECTs with cross-linked P3HT are minimal even after 100 cycles of operation; moreover, the cross-linked OECTs exhibit faster switching properties. In addition, the enhanced doping efficiency allows for the clear observation of plasmon resonances in nanostructured, highly doped P3HT. We believe that the proposed technique for internalizing ionic transport abilities can be applied to various ion-based semiconductor applications.



KEYWORDS: *cross-linking, electrochemical doping, ion-gating transistors, nanoantennas, localized surface plasmon resonance (LSPR)*

INTRODUCTION

Interest in ionic transport within organic semiconductors is rapidly growing, particularly in fields of research such as organic electrochemical transistors (OECTs),[1,2] biosensing applications,[3,4] synaptic devices,[5,6] and switchable nanoantennas.[7−10] The reversible control of the electrical and optical properties of organic semiconductors through the injection or extraction of ions via an applied voltage (i.e., the electrochemical doping process[11]) presents a compelling mechanism that conventional oxide- or silicon-based semiconductors struggle to implement. This has consequently shifted research focus toward managing not only charge transport but also the ionic mobility of active materials in the field of organic electronics/optics. Conventional polymer semiconductors like poly(3-hexylthiophene) (P3HT) exhibit excellent charge transport properties due to their high crystallinity. However, their inherently hydrophobic nature significantly hinders aqueous doping and hydrated ion uptake/release processes. In particular, these materials are not adequate for applications involving ionic doping with aqueous salts or polar ionic liquids, as their low ion transport capabilities result in reduced doping efficiency and slow transition speeds.[12,13] Recently, material modification studies, such as those involving the introduction of polar side chains to increase the ionic conductivity of these hydrophobic polymer semiconductor materials, have been conducted;[14−16] however, these material modification techniques yield undesirable effects, such as altering the inherent charge transport properties of the polymeric material, which prevent their application as universal methods.[17,18]

Electrochemical doping has also been adopted in switchable nanoantenna technology research, an emerging area within metasurface research.[7] Conductive polymeric materials can be excellent candidates for such research due to their controllable optical properties. For instance, nanostructured poly(3,4-

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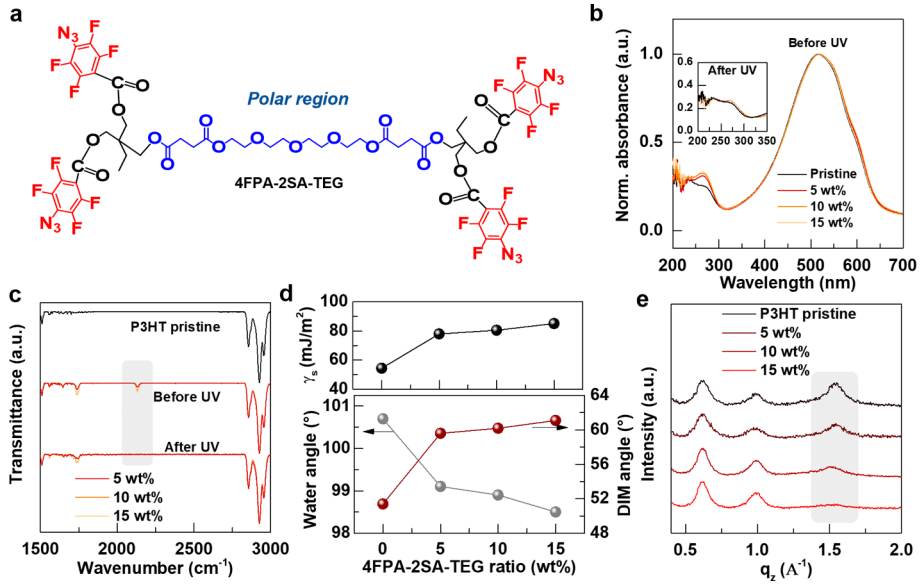


Figure 1. (a) Chemical structure of the synthesized cross-linker. (b) UV−vis and (c) FT-IR spectra of simply mixed P3HT:4FPA-2SA-TEG films with respect to the weight ratio of the blend film. The inset of (b) highlights the spectra after photo-cross-linking. (d) Water and diiodomethane contact angle plots for thin films with varying cross-linker weight ratios. Surface energy, calculated using the Owens−Wendt model based on the contact angle data, is also presented. (e) Out-of-plane line-cut plots of corresponding 2D GIXD patterns with various ratios of the cross-linker.

ethylenedioxythiophene) (PEDOT)-based highly conductive polymers can act as optical nanoantennas that interact strongly with light, thereby supporting localized surface plasmonic resonances (LSPRs) in the shortwave infrared (SWIR) region.[7−10] Owing to the tunable electrical conductivity and metallic characteristics afforded to these polymers by electrochemical doping, LSPRs can be electrically switched on or off. However, PEDOT-based polymers generally exhibit slow optical transition during electrochemical doping (≈0.02 Hz with solid electrolyte[9]) because of their poor ionic charge transport behavior, which has been identified as the primary speed-limiting factor.[19,20] Additionally, their low ionization potential (typically <5.0 eV[21]) results in the formation of byproducts (e.g., H_2O_2) during repeated electrochemical doping processes, leading to rapid degradation of the devices.[15,21] Thus, it is essential to design polymer semiconductor systems that can offer high ion mobility and operational stability without sacrificing charge carrier mobility.

Herein, we propose a photochemical approach to universally internalize ionic transport properties within hydrophobic polymer semiconductors. We developed a cross-linker that significantly enhanced the ion transport ability of P3HT, known to be incompatible with electrochemical doping processes due to its hydrophobic nature. This cross-linker featured four azide photophores capable of photochemically bonding with the alkyl chains of polymers, particularly those incorporating long ethylene glycol bridges, which are known for their high polarity between photophores. This cross-linker had the following features: (1) micropatterning capabilities, (2) ability to establish ion transport channels within the polymer through its polar regions, and (3) highly improved electrochemical doping stability. Furthermore, we confirmed that the polymer semiconductor was homogeneously cross-linked, even at a low cross-linker concentration (5 wt %), and quantitatively analyzed the effects of incorporating the cross-linker by applying various measurement tools, including electrochemical impedance spectroscopy (EIS).

We also compared device performance with and without cross-linkers. Interestingly, the formation of an ion transport channel within the polymer significantly enhanced the operational stability of OECTs. Particularly, the transfer curves confirmed that the on/off ratio and saturation mobility did not decrease, even after 100 cycles, whereas significant degradation in operational stability was observed for the untreated sample. Additionally, we demonstrated nanostructured polymer metasurfaces to assess their LSPR switching capabilities. We obviously confirmed the electrical switchability of these nanoantennas by observing a voltage-dependent plasmon resonance peak appeared in the SWIR region following electrochemical doping. Notably, a linear increasement in resonance amplitude and faster optical transition speed were observed for the cross-linked samples than for the untreated samples; this was attributed to their efficient ionic doping capabilities. These findings highlight the potential for organic-based electrically tunable metasurface research to expand beyond PEDOT, incorporating polymer semiconductors.

RESULTS AND DISCUSSION

Introduction of Functional Cross-Linkers. We synthesized four tetrafluorophenyl azide (FPA)-based photochemical cross-linkers, each with a tetraethylene glycol (TEG) bridge. Their chemical structures are shown in Figure 1a, and detailed synthetic procedures and characterizations are outlined in Scheme S1 and Supporting Method 1. The FPA group is a well-known moiety that efficiently forms bonds when exposed to UV light, particularly with secondary carbons ($-CH_2$) of alkyl chains, thereby enabling cross-linking with various materials.[22] Building on our previous studies on the influence of TEG groups on the hydrophilicity of cross-linked systems,[23] we designed cross-linkers with the expectation that the TEG bridges would effectively provide ion transport channels within the system. To verify that cross-linking occurred within the polymer, we examined the UV−vis spectra of the P3HT films before and after photo-cross-linking (Figure 1b). The peaks of

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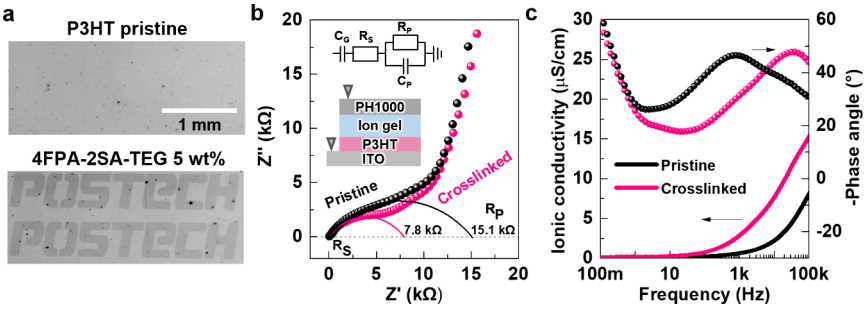


Figure 2. (a) Optical microscopy images of photopatterned P3HT and cross-linked films. The POSTECH logo was reproduced with permission from Pohang University of Science and Technology. (b) Nyquist and (c) Bode plots extracted from EIS measurements obtained in the presence or absence of cross-linking; 5 wt % cross-linkers were used for comparison with the pristine sample. The inset of (b) shows the equivalent circuit and schematic of the device applied for the measurements. V_{AC} was applied with a 25 mV amplitude, whereas V_{DC} was set to the open circuit potential of the device. The $[EMIM]^+[TFSI]^-$-based solid-state gel electrolyte and PH1000-based top electrode are used for the evaluation.

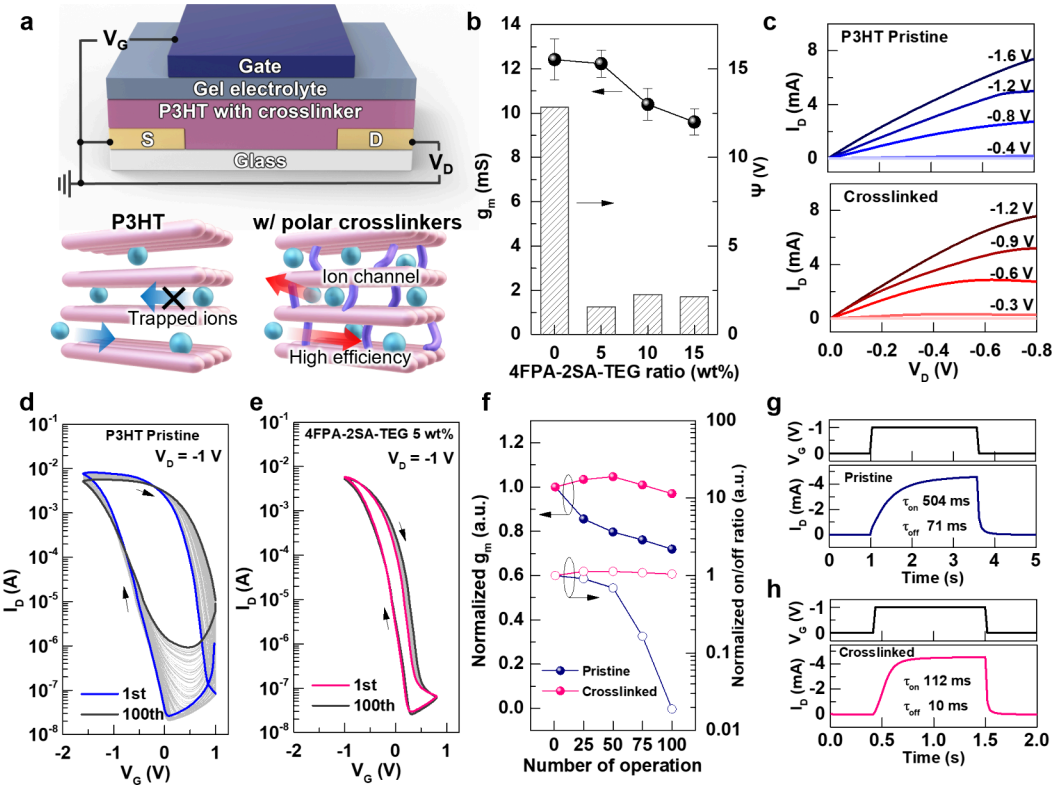


Figure 3. (a) Schematic of the fabricated ion-gating transistors applied for the measurements. The $[EMIM]^+[TFSI]^-$-based solid-state gel electrolyte and PH1000-based top electrode are used for the evaluation. (b) Variation in transconductance and hysteresis strength as a function of cross-linker concentration, with error bars representing the standard deviation for six devices. (c) Output characteristics of ion-gating transistors; V_G was varied from 0 to −1.6 V for the pristine film and from 0 to −1.2 V for the cross-linked (5 wt %) film. Measured I_D-V_G characteristics over 100 cycles (d) for the pristine P3HT film and (e) for the cross-linked (5 wt %) film. V_D was set to −1 V for both film types, with V_G ranging from 1 to −1.5 V for the pristine film and 0.8 to −1 V for the cross-linked (5 wt %) film. The scan rate was 25 mV s^{-1}. (f) Plots of normalized transconductance and normalized on/off ratio, as derived from the results shown in (d) and (e). Transient curves of drain current for (g) pristine and (h) cross-linked (5 wt %) films under a gate pulse of −1 V, with V_D held constant at −1 V. Time constants are extracted by fitting the exponential decay function.

the FPA moiety in the range of 250−300 nm were subsequently reduced after 254 nm UV light irradiation, without affecting original absorption spectrum of P3HT films. As depicted in Figure 1c, the Fourier-transform infrared (FT-IR) spectra confirm the presence of azide stretching bands characterized by an N=N−N stretching frequency (~2100 cm^{-1}) in the P3HT-containing cross-linkers before photo-cross-linking, whereas no distinct bands corresponding to this frequency are observed for the pristine system. Following 254 nm UV light irradiation, the azide stretching band for the photo-cross-linked P3HT disappears due to cross-linking regardless of the concentration of the cross-linkers, with no changes observed in P3HT's original peaks. Analysis of the UV−vis and IR spectra of the P3HT thin films confirm successful cross-linking within the system.

The contact angle measurement results for water and diiodomethane (DIM), presented in Figure 1d and Figure S1, demonstrate that the polar cross-linker effectively adjusts the

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polarity of thin films. With 5 wt % cross-linker, the water contact angle decreased from 100.7 to 99.1°, while the DIM contact angle increased from 51.3 to 59.6°. The calculated surface energy showed a clear dependence on cross-linker concentration, with 5 wt % cross-linker significantly increasing the surface energy from 54.4 to 78.0 mJ m^{-2}. At higher cross-linker concentrations, changes in contact angle and surface energy became more gradual. The GIXD and its out-of-plane line-cut analysis (Figure 1e and Figure S2) revealed that as the proportion of the cross-linker increases, the intensity of the out-of-plane $\pi-\pi$ stacking peak weakens, and it is almost diminished when the cross-linker content exceeds 15 wt %. This indicates that excessive cross-linker significantly disrupts the intrinsic $\pi-\pi$ stacking of P3HT. Therefore, using the minimum amount of cross-linker is essential to maintaining high charge mobility of the OECT.

Functionality of Polar Cross-Linkers. The most important effect that can be expected from incorporating a photo-cross-linker into a polymer semiconductor thin film is photopatterning.[16,24] High-resolution patterning is crucial to simplify the fabrication of advanced electronics such as memory, synaptic devices, and logic circuits. Thin P3HT:4FPA-2SA-TEG films were irradiated with 254 nm UV light using shadow masks, followed by rinsing with chloroform. As shown in the optical microscopy images in Figure 2a, the pristine P3HT film is erased upon development, as expected. However, successful patterning is achieved with the addition of 5 wt % cross-linker. Notably, as shown in Figure S3a,b, there are no significant differences in the optical microscopy images and film thicknesses after patterning with varying concentrations of cross-linkers.

EIS was conducted to verify the ion transport properties of the photo-cross-linked system. An ionic gel was used to evaluate the properties of the all-solid-state electrochemical systems. To promote the effective formation of electric double layers (EDLs) within the device, we utilized PEDOT as the top electrode. PEDOT is known for its ability to efficiently accumulate ions at the interface because of its relatively high specific capacitance.[25]Figure 2b shows the Nyquist plots derived from EIS measurements; as anticipated, no significant difference is observed in the electrolyte resistance (R_S) of pristine and cross-linked films. However, the estimated charge transfer resistance (R_P) decreases from 15.1 to 7.8 kΩ with the addition of 5 wt % cross-linkers. Because R_P indicates the ease of ion transfer across the P3HT/ion gel interface, this reduction suggests that the polar nature of the TEG bridge in the cross-linkers effectively establishes an ionic pathway throughout the P3HT film. As can be ascertained from Figure 2c, the Bode plot provides more intuitive results. The ionic conductivity increases with the addition of cross-linkers, nearly doubling in the whole frequency range (8.0 to 15.2 μS cm^{-1} for 100 kHz). Notably, the peak of the phase angle shifts toward the high-frequency region as a result of cross-linking, whereas the intensity decreases in the intermediate-frequency region (approximately 1−1000 Hz).[26] These shifts indicate that the ionic pathways established by the cross-linker enhance the capability of the device to form an EDL at higher frequencies, facilitating more active and faster ionic diffusion in the intermediate-frequency region. As shown in Figure S4, cyclic voltammetry (CV) measurements demonstrate a higher current density in the cross-linked film, exhibiting a clearer and more linear dependence of current on the scan rate. This behavior suggests efficient and stable counterion penetration across the bulk of the cross-linked film.[27] EIS and CV results indicate that the ionic transport properties of the P3HT film can be effectively enhanced by using TEG-bridged cross-linkers.

Charge Transport Characteristics in Ion-Gating Transistor Systems. To evaluate the effects of the cross-linker in the systems, we initially fabricated OECTs; the schematic is shown in Figure 3a and the digital image of OECT device is shown in Figure S5. Owing to their ion-gating mechanism, OECTs are well suited to analyze properties such as the ionic transport and operational stability of polymer semiconductors. Figure 3b shows the transconductance (g_m) and hysteresis strength (Ψ) of systems determined from the I_D-V_G curves shown in Figure S6 and the following equations for varying cross-linker concentrations:[28]

$$g_m = \frac{\partial I_D}{\partial V_G} = \mu C^* \left(\frac{Wd}{L}\right)\left(V_G - V_{th}\right) \quad (1)$$

$$\Psi = \int \theta(V_G)\, dV_G \quad (2)$$

$$\theta(V_G) = \ln\left(\frac{I_D(t_2)}{I_D(t_1)}\right) \text{ when } V_G(t_1) = V_G(t_2) \quad (3)$$

where μ is the charge carrier mobility, C^* is the volumetric capacitance of the polymer, L is the channel length, d is the channel thickness, W is the channel width, V_{th} is the threshold voltage, and $I_D(t_1)$ and $I_D(t_2)$ are the I_D for the same V_G during switching on and off, respectively. A significant decrease is observed in the hysteresis strength from 12.9 to 1.6 V in response to the 5 wt % cross-linker, indicating more stable, predictable, and faster ionic doping processes over repeated operations. The transconductance is not significantly different between the pristine film and the 5 wt % cross-linked film (about 12 mS); however, a rapid decrease is observed when cross-linkers are applied at more than 10 wt %. This suggests that an excessive amount of cross-linkers can disrupt charge transport in the polymer thin film, allowing significant decrease in μ. The output characteristics of the OECTs are illustrated in Figure 3c. Compared with the pristine system, the OECT with cross-linked P3HT (5 wt %) exhibits a well-saturated current at lower V_G and V_D values, suggesting that the electrochemical doping process enhances the ion transport properties and consequently increases the power efficiency.

To compare the operational stability of the devices, transfer curves were measured over 100 cycles for pristine and cross-linked P3HT (Figure 3d,e). A substantial increase (i.e., by a factor of 50) in the off-current is observed for the pristine P3HT; this increase has been attributed to the permanent doping of the channel with irreversibly trapped anions.[29] Additionally, the on-current is halved as a result of the repeated external stress application impeding charge transport. Interestingly, for the cross-linked P3HT, both the on- and off-currents remain constant during repetitive device operations, with only a slight increase in hysteresis. This suggests that the enhanced ionic transport within the polymer prevents undesirable ion trapping, which is the main contributor to the degradation of ion-gating systems.

To quantify the stability of the systems, the normalized transconductance and on/off ratio plots were constructed. As shown in Figure 3f, for the pristine P3HT, both parameters significantly decrease under repeated operation, reaching 69%

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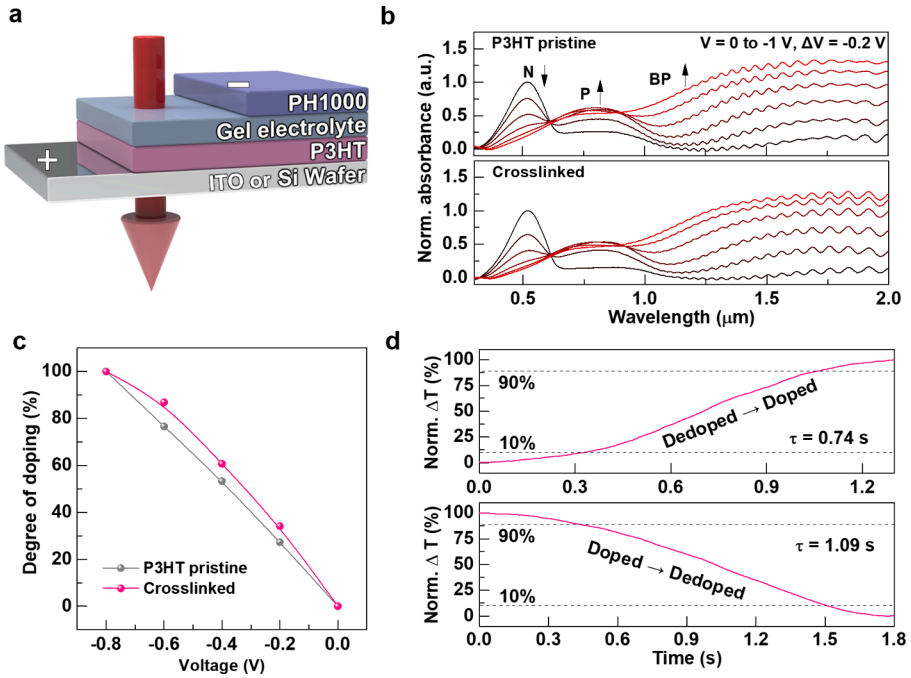


Figure 4. (a) Schematic of the device fabricated for optical measurements, with the red arrow indicating the path of the electromagnetic wave. (b) UV−vis−NIR spectroscopy results for the pristine and cross-linked P3HT film on an ITO substrate under various applied voltages. (c) Plots showing the degree of doping as a function of the applied voltage, calculated from the analysis of UV−vis−NIR spectra. (d) Transient curves for normalized transmission of the system. Voltages of −1 and 0 V were applied to measure the rise time associated with neutral P3HT doping and fall time associated with the doped P3HT returning to its neutral state, respectively.

and 1.7% of the initial values for saturation mobility and on/off ratio, respectively. In contrast, for the cross-linked P3HT, no significant difference is observed in the on/off ratio and the transconductance is even improved over 50th cycle; this is attributed to a stabilization during repeated electrochemical doping processes. Cycling tests with continuous gate pulses, as shown in Figure S8, further support this observation. While the pristine P3HT exhibited an increased off-current at higher cycle numbers, the cross-linked P3HT demonstrated a decreased off-current, highlighting its enhanced stability.

Regarding comparison of the driving speeds, Figure 3g,h presents the transient drain current with a gate pulse. While the pristine P3HT exhibits time constants of 504 and 71 ms for on and off switching, respectively, the cross-linked P3HT demonstrates significantly faster switching, with time constants of 112 and 10 ms for on and off states, respectively. This result is consistent with the EIS measurements, indicating that more active and faster ion transport occurs in the cross-linked P3HT film. The OECT measurement results confirm that the cross-linker effectively enhances the efficiency, stability, and dynamics of the ionic doping processes without compromising the charge transport properties of the polymer.

To evaluate the volumetric doping characteristics, transconductance measurements were conducted for pristine and cross-linked films with varying thicknesses, as shown in Figure S9. Both pristine and cross-linked films exhibited strong linearity with increasing film thickness, confirming three-dimensional ion injection and charge transport.

Figure S10 compares the performance of previously reported OECT channel materials (PEDOT:PSS and p($g_3$2T-T)) by measuring I_D−V_G characteristics and transconductance. Since a direct comparison with prior studies may not account for differences in device geometries and electrolyte materials, we carefully measured and evaluated normalized transconductance values ($g_m/(Wd/L)$) using the identical electrolytes and device geometry. Notably, cross-linked P3HT demonstrated the highest transconductance among the materials tested, even surpassing p($g_3$2T-T), which is considered one of the state-of-the-art OECT materials.[5,14] This finding highlights cross-linked P3HT as a promising candidate for ion-gel-based applications, offering both high stability and excellent transconductance.

Electrochemically Tunable Optical Properties of Films. Prior to fabricating the nanoantennas and assessing the LSPR switchability using the cross-linked P3HT, we examined the optical properties of the thin films. Electrochemical doping offers significant geometric freedom because the electrode only needs to contact a part of the electrolyte.[2] This allowed us to design a system structure (Figure 4a) where electromagnetic waves can pass through in the absence of interference from the top electrode. Figure 4b shows the change in absorbance in the visible-NIR region for pristine and cross-linked P3HT films. When a negative voltage is applied, ionic doping causes the neutral P3HT peak (~520 nm) to disappear and a peak corresponding to polaron (~800 nm) and bipolaron absorption (>1000 nm) to appear, which is consistent with previous reports on doped P3HT.[30,31] To evaluate the effect of cross-linking on the degree of doping, multipeak fitting was performed on the UV−vis-NIR spectra, as shown in Figure S11a. The ratio of neutral species (N_n) was determined by calculating the area of absorption spectra in the neutral region (Figure S11b). These values were then used to calculate the degree of doping using the formula; $1\text{-}N_n/N_{n,0}$. The degree of doping is plotted in Figure 4c as a function of the applied voltage, and the cross-linked P3HT film exhibited enhanced efficiency in electrochemical doping compared to the pristine one. This finding aligns with the OECT output characteristics discussed earlier, where the cross-linked film

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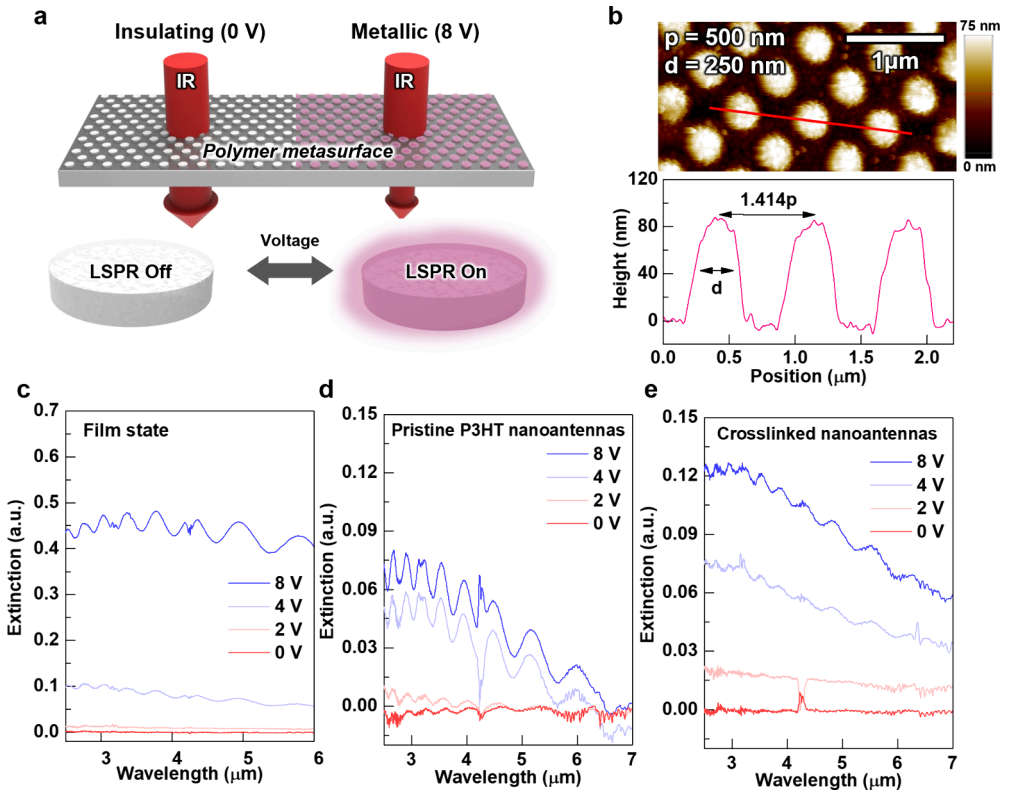


Figure 5. (a) Schematic illustrating the driving mechanism of switchable nanoantennas. (b) AFM images of the nanopatterned P3HT film, with each disk having a diameter of 250 nm and a period of 500 nm. (c) Measured FT-IR spectra for the cross-linked P3HT film on a Si wafer substrate under various applied voltages. Measured extinction spectra for (d) pristine and (e) cross-linked nanoantennas as a function of applied voltage. The spectral activity at 4.3 μm is attributable to absorption by CO_2.

demonstrated improved power efficiency in electrical transitions.

It is reasonable to assume that the driving speed of switchable nanoantennas corresponds to the optical transition speed during the doping and dedoping processes of the film. A 532 nm laser diode was used to measure the transition speed of the transmittance of the device (Figure 4d). A voltage of −1 V was applied to the top electrode to dope the P3HT film. When the applied voltage is removed (0 V), the anions injected by the external voltage are released into the gel, allowing the doped P3HT film to return to its neutral state.[32] The rise time associated with the transition in transmittance during P3HT doping is 0.74 s, and the fall time associated with the transition in transmittance during dedoping is 1.09 s, indicating that the nanoantenna with cross-linked P3HT has a switching speed of approximately 1 Hz. Optical transitions exhibit slower dynamics compared to electrical transitions observed in OECTs (~112 ms), primarily due to the faster doping dynamics in the ordered regions compared to the disordered regions within the polymer film.[33] Electrical transitions occur immediately upon the formation or disruption of the electrical percolation path, which corresponds to the doping or dedoping of the ordered regions. In contrast, optical transitions require both the ordered and disordered regions to be fully doped or dedoped to achieve saturation. The rise and fall times for the pristine P3HT are 1.67 and 3.59 s (Figure S12), demonstrating that the enhanced ionic conductivity of the cross-linker facilitates a faster optical transition. These results are consistent with the previously discussed EIS analysis and OECT operational speed measurements. Notably, this optical transition speed is significantly higher than that of PEDOT-based devices[9] (0.02 Hz) using identical solid-state electrolytes. This is because substantially more time is required to dedope highly doped PEDOT since PEDOT is not inherently ionically conductive.[17,18]

LSPR-Switchable Nanoantennas. A schematic of the LSPR-switchable nanoantenna is shown in Figure 5a. Polymer semiconductors are optically insulating when they have a low charge carrier density (dedoped). However, when electrochemically doped by applying positive voltages, they become optically metallic with a sufficiently high charge carrier density, shifting the plasma wavelength to the observable spectral range. When nanostructuring is applied, the surface plasmons become confined in the nanostructures, which is known as LSPRs.[7] The obtained plasmon resonance is switchable because of reversible ionic doping processes that effectively control the charge carrier density. To realize LSPR for polymer semiconductors, disk-patterned nanoantennas were fabricated using P3HT. A detailed description of nanoantenna fabrication is provided in the Methods section. Figure 5b shows an atomic force microscopy (AFM) image and the height distribution of the nanoantennas. The diameter and periodicity of the nanodisks are 250 and 500 nm, respectively. The AFM image confirms the successful realization of nanoantenna arrays, with the height of each antenna fixed at 80 nm.

Figure 5c shows the FT-IR spectra for cross-linked P3HT film under different applied voltages. A relatively higher voltage was required compared to Figure 4 which utilizes an ITO substrate (~10^{-4} Ωcm), because the Si wafer substrate used in this study exhibits higher resistivity of 10^{1} Ωcm. As the voltage increases from 2 to 8 V, the degree of doping increases, resulting in flat and broad absorption across the entire SWIR

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region. This is indicative of clear bipolaron absorption in highly doped P3HT,[34] exhibiting an intensity similar to that of the PEDOT film measured in our previous study.[10] This also implies that electrochemically doped P3HT may achieve sufficiently high charge carrier density and thus conductivity to excite surface plasmons. The plasma frequency (ω_p) and plasma wavelength (λ_p), the lower bound of the plasmonic spectral region, can be defined as follows:

$$\omega_p = \sqrt{\frac{ne^2}{\varepsilon_0 m_e}} \tag{5}$$

$$\lambda_p = \frac{2\pi c}{\omega_p} \tag{6}$$

where n is the charge carrier density, c is the speed of light, m_e is the effective mass of the charge carrier, and ε_0 is the permittivity of free space. Indeed, the estimated plasma frequency and wavelength are highly dependent on the charge carrier density, which implies that a sufficiently high number of charge carriers is required to excite the surface plasmons in the observable spectral range. Given that previous studies on nanoantennas using PEDOT[7,10] or chemically doped polymer semiconductor[34] have reported plasmon resonances in the range of 2 to 6 μm, we anticipate that the LSPRs for highly doped P3HT-based nanoantennas will have a similar range. It is worth mentioning that besides the charge carrier density, the mobility[35] and effective mass[36] of the charge carrier may also be affected by electrochemical doping. However, all those changes induced by doping will lead to the shift of the plasma frequency/wavelength in the same direction, and the contribution from the charge carrier density would be the most dominant among them.

As shown in Figure 5d,e, IR spectra for pristine and cross-linked P3HT-based nanoantennas exhibit flat absorption behavior when low voltages are applied, similar to those for the film-state P3HT (Figure 5c); this implies that the degree of doping is insufficient to achieve LSPRs. However, as the P3HT nanoantennas are highly doped with high applied voltages, clear resonance features are successfully obtained with the peak position at a wavelength of 3 μm. Notably, the LSPRs for the cross-linked P3HT become visible at lower voltages and the peak intensity almost linearly increases with increasing voltages up to 8 V, whereas the LSPRs for the pristine P3HT show a weak saturation for high voltages. This suggests that more efficient doping occurred in the cross-linked P3HT, enabling LSPR activation at a wider range of voltages. This improvement is attributable to the increased ionic mobility of P3HT, which enhances doping efficiency and electrochemical stability, as previously discussed. The functional cross-linker used in this study enabled the polymer semiconductor to internalize the ionic conductivity, showcasing the potential for organic dynamic metasurface research to move beyond PEDOT, incorporating polymer semiconductors. Looking forward, we believe that achieving ultrahigh doping levels in organic polymers and a further shift of the plasmonic resonance may find use in a variety of applications in the visible range in the near future.

CONCLUSIONS

In this paper, we demonstrated the successful internalization of ion transport channels within polymer semiconductors using an FPA-based photochemical cross-linker with a TEG bridge. This cross-linker significantly enhanced the electrochemical stability, ionic mobility, and doping efficiency of P3HT, making it suitable for use in high-performance OECTs and electrochemically switchable nanoantennas. We confirmed the ionic conductivity in cross-linked P3HT films to be twice that in pristine films, and that the performance degradation due to undesired ion trapping was effectively prevented in the cross-linked films. Our results also indicated that cross-linked P3HT was able to achieve faster electrical and optical transition speeds than those of pristine P3HT while preserving its charge transport properties. These improvements are crucial for the development of ion-based advanced organic electronic/optic devices. The photochemistry-based ion transport property internalization technique proposed in this paper will pave the way for the precise control of the ion transport and charge transport properties of various semiconductor materials. In this regard, this study has demonstrated that polymer semiconductors, in addition to highly conductive materials, can be effectively incorporated into the candidate materials for electrically tunable dynamic metasurfaces. We believe that the application of polymer semiconductors will soon lead to the development of switchable nanoantennas capable of controlling visible light, bringing this advancement closer to reality.

METHODS

Materials. 4-(Dimethylamino)pyridine (DMAP), TEG, trimethylolpropane, and succinic anhydride were procured from TCI. *N*,*N*′-dicyclohexylcarbodiimide (1.0 M in methylene chloride), $CDCl_3$, and anhydrous dichloromethane were obtained from Sigma-Aldrich. $MgSO_4$, *n*-hexane, ethyl acetate, and diethyl ether were purchased from Daejung Chemicals Korea. Additionally, 4-azido-2,3,5,6-tetrafluorobenzoic acid was synthesized in our laboratory according to a previously published method.[20] P3HT was purchased from Ossila. Poly(vinylidene fluoride-*co*-hexafluoropropylene) (P(VDF-HFP)), 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)-imide ([EMIM][TFSI]), and ethylene glycol were purchased from Sigma-Aldrich. Poly(3,4-ethylenedioxythiophene) polystyrenesulfonate (PEDOT:PSS; PH1000) was purchased from Heraeus. The materials were used as received without further purification.

Synthesis Materials and Characterization. 4FPA-2SA-TEG was synthesized by following the procedure described in Supporting Methods and Scheme S1. ^{1}H, ^{13}C, and ^{19}F NMR spectra were recorded by applying a Bruker Avance-400 spectrometer using tetramethylsilane as an internal standard. Chemical shifts are reported in parts per million (ppm).

Device Fabrication. ITO and glass substrates were cleaned sequentially by sonication in a Mucasol solution, distilled water, acetone, and isopropyl alcohol for 10 min each. Double-sided polished (DSP) Si wafer substrates were cleaned with a piranha solution. The substrates were then dried under N_2 flow. For the electrodes used in ion-gating transistors, Ti (3 nm) and Au (37 nm) were deposited via thermal evaporation. For the active layer, P3HT:4FPA-2SA-TEG (100:0, 5, 10, 15 w/w) were dissolved in chloroform at concentrations of 10 mg mL^{-1}. The solution was stirred at 50 °C for 12 h, then spin-coated at 1500 rpm on the substrate. Before annealing the film at 80 °C for 20 min, 254 nm UV irradiation was applied to the samples for 10 min to facilitate photo-cross-linking. For the solid-state electrolyte, P(VDF-HFP):[EMIM][TFSI]:acetone (1:4:7 w/w/w) was stirred at 70 °C for 6 h. The solution was spin-coated at 1500 rpm onto a washed glass substrate and dried at room temperature to form a freestanding film. The solid electrolyte film was cut into pieces using a razor blade and transferred to the desired position using tweezers. For the PH1000 top electrode, a freestanding film of PEDOT:PSS formed on a Si wafer via drop casting was cut and transferred to the desired position. To increase the electrical conductivity of the gate electrode, ethylene glycol (10 vol %) was added to the PH1000 solution before deposition.

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Thin-Film Characterization. A Thermo Scientific Evolution 220 UV−vis spectrophotometer and an Ocean Insight HDX-VIS-NIR system were used to obtain the UV−vis-NIR absorption spectra. FT-IR spectroscopy (Vertex 70v, Bruker) was used to confirm the photochemical reactions of the cross-linkers. The GIXD measurements were performed at the Pohang Accelerator Laboratory (PAL) in the Republic of Korea. The 3C and 9A beamlines were used for measurement.

Transistor Characterization. The electrical characteristics of the transistors, including the transfer and output curves, were measured at room temperature in a N_2 glovebox using a Keysight B1500A semiconductor parameter analyzer. EIS measurements were performed using a potentiostat (AUTOLAB PGSTAT302N, Metrohm). The transient response was measured using a source-meter unit (Keithley 2636B).

Nanoantenna Fabrication. The polymer solution was prepared and spin-coated onto the precleaned DSP Si wafer following the same procedure described in the Device Fabrication Section. The preparation method for the Si master template and a schematic of the nanopatterning method are shown in Supporting Method 2 and Scheme S2, respectively. Before each replication process, a hydrophobic treatment was applied to the original template or mold by exposing the surface to vapors of 0.03 M trichloro(1H,1H,2H,2H-perfluorooctyl)silane (FTS, Sigma-Aldrich) dissolved in hexane for 1 h, followed by annealing at 120 °C for 20 min. The first mold was prepared on a glass substrate. To maximize adhesion, the glass substrates were covered with a glass primer (Changsung Sheet) by applying spin coating at 3000 rpm for 40 s, followed by baking at 135 °C for 10−20 min. Then, 1−2 drops of a polyurethane acrylate (PUA) resin (311RM, Changsung Sheet) were deposited onto a Si master template and covered with a glass substrate. Following UV curing and the removal of excess resin, the glass substrate with the nanopatterned resin was detached from the Si master, yielding the first mold. The second molds were prepared by repeating the same replication process using a more flexible PUA resin (ERM, Changsung Sheet) and polydimethylsiloxane substrates. A photoresist (AZ5214E, diluted with propylene glycol monomethyl ether acetate at a 1:5 volume ratio, MicroChemicals) was cast on the second mold and directly sandwiched between the mold and P3HT films (both pristine and cross-linked). After drying at 50 °C for 30 min, subsequent cooling to room temperature, and mold detachment, the photoresist nanopatterns were left on the P3HT film. Reactive oxygen plasma etching (16 sccm, 100 W) was performed using the photoresist nanopatterns as a mask. To confirm the complete etching of the exposed parts of the P3HT film that were not covered by a photoresist, multiple rounds of etching (~20 s) were performed. After fabricating the nanostructures of both pristine and cross-linked P3HT, the solid-state electrolyte and PEDOT:PSS top electrode were precisely transferred to the designated locations, forming the electrode/electrolyte/nanoantennas/wafer structure. This process followed the procedure that outlined in the Device Fabrication section.

Nanoantenna Characterization. AFM (NX10, Park systems) was used to observe the nanoantenna structure. Electromagnetic waves were irradiated adjacent to the top electrode, following a setup identical to the UV−vis-NIR measurement illustrated in Figure 4a. Bruker HYPERION II was used to obtain FT-IR spectra for thin films and nanoantennas. Keithley 2450 SMU was used to apply voltage to the device. An oscilloscope (GDS-2102A, GW Instek), a reference photodetector (APD10A2, Thorlabs), and a 532 nm laser diode (CPS532, Thorlabs) were used to evaluate the transient response of the device transmittance.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsnano.4c18087.

Synthesis method of 4FPA-2SA-TEG (Supporting Method 1); fabrication method of Si master templates (Supporting Method 2); synthesis routes for 4FPA-2SA-TEG (Scheme S1); contact angle measurements for various film conditions (Figure S1); 2D GIXD measurement patterns for various film conditions (Figure S2); direct photopatterning results of P3HT:4FPA-2SA-TEG films and film thickness variations after rinsing (Figure S3); CV measurements for pristine and cross-linked films with various scan rates (Figure S4); digital image of the OECT device (Figure S5); transfer curves of OECTs with various cross-linker concentrations (Figure S6); measured gate current over 100 cycles for pristine and cross-linked P3HT-based OECTs (Figure S7); measured drain currents with continuous gate pulses for pristine and cross-linked P3HT-based OECTs (Figure S8); transconductance plots with various device geometries (Figure S9); transfer curves and transconductance plots of OECTs with various channel materials (Figure S10); multipeak fitting for UV−vis−NIR measurements and normalized neutral species density plots (Figure S11); measured transient photonic response of the P3HT thin film (Figure S12); fabrication processes for nanostructured P3HT (Scheme S2) (PDF)

AUTHOR INFORMATION

Corresponding Authors

Evan S. H. Kang − *Department of Physics, Chungbuk National University, Cheongju 28644, Republic of Korea*; Email: eshkang@chungbuk.ac.kr

Dae Sung Chung − *Department of Chemical Engineering, Pohang University of Science and Technology (POSTECH), Pohang 37673, Republic of Korea*; orcid.org/0000-0003-1313-8298; Email: dchung@postech.ac.kr

Authors

Won Jun Pyo − *Department of Chemical Engineering, Pohang University of Science and Technology (POSTECH), Pohang 37673, Republic of Korea*

Seunghyun Lee − *Department of Physics, Chungbuk National University, Cheongju 28644, Republic of Korea*

Syed Zahid Hassan − *Department of Chemical Engineering, Pohang University of Science and Technology (POSTECH), Pohang 37673, Republic of Korea*; orcid.org/0000-0001-7508-4905

Dowan Kim − *Department of Chemical Engineering, Pohang University of Science and Technology (POSTECH), Pohang 37673, Republic of Korea*; orcid.org/0000-0001-9759-907X

Junho Jung − *Department of Physics, Chungbuk National University, Cheongju 28644, Republic of Korea*

Complete contact information is available at: https://pubs.acs.org/10.1021/acsnano.4c18087

Author Contributions

W.J.P. and S.L. carried out most of the experimental work and data analysis. They contributed equally to this work. S.Z.H. provided materials and assisted with the data analyses. D.K. and J.J. assisted with the visualization and methodologies. D.S.C. and E.S.H.K. supervised the work. All authors have contributed to the interpretation of the results and the preparation of the manuscript, which was drafted by W.J.P.

Notes

The authors declare no competing financial interest.

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